

Not only tech giants like Intel are building ML processors, even startups are building these now. For instance, Cerebras Systems has developed an ML processor with 400,000 cores and 18GB SRAM using wafer-scale integration technique instead of the traditional process of making multiple chips from a single wafer for low latency and high bandwidth. For high yields, it had to overcome the design challenges, and redundant processor cores had to be incorporated to deal with chip imperfections.

Open source hardware

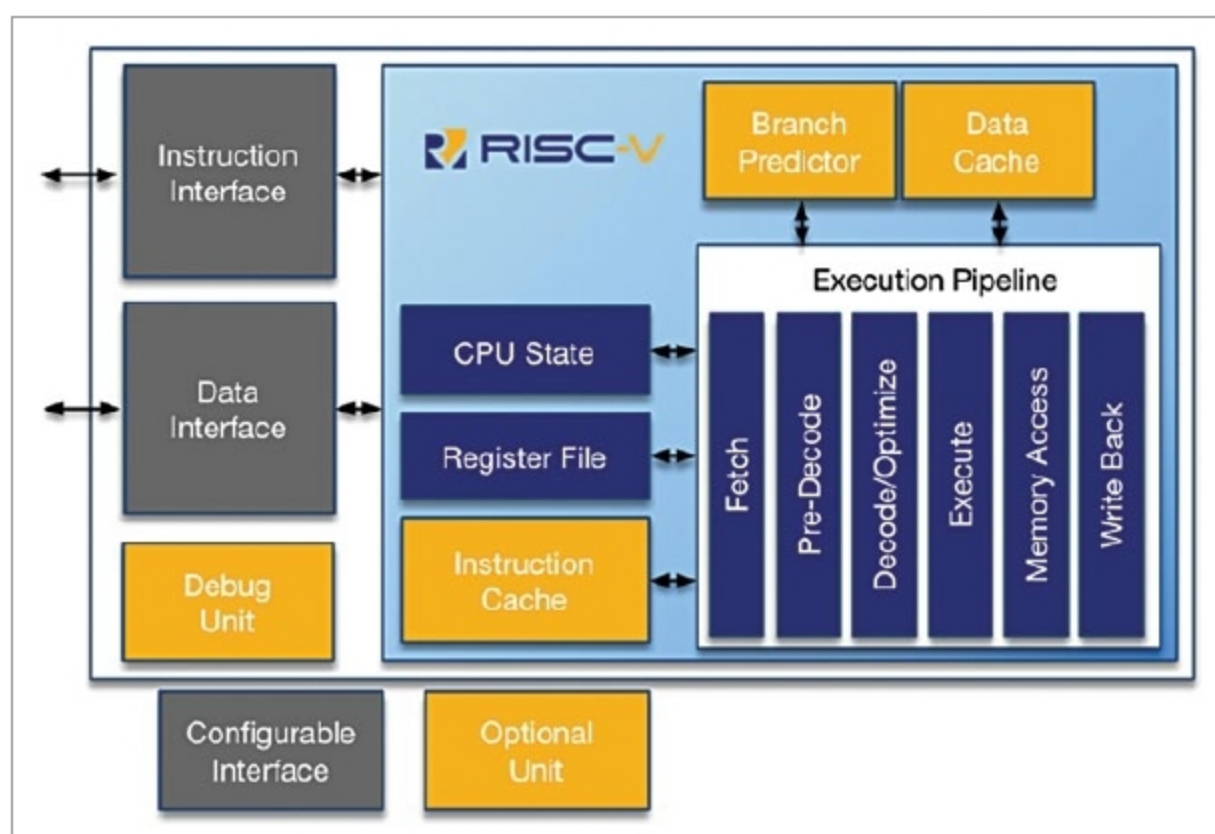
To achieve a common standard and aid the growing community of designers, RISC-V (reduced instruction set computer) is the latest open source hardware architecture that can be tailored to suit various application requirements. Unlike ARM, it is more flexible and is useful for maximum dissemination, thereby expanding the market and ecosystem.

Companies such as Google, Nvidia, Qualcomm and Western Digital are supporting RISC-V. In June, X-FAB Silicon Foundries, together with Efabless Corp., announced the silicon availability of Raven, an open source RISC-V microcontroller (MCU). Ulrich Bretthauer, product marketing manager, X-FAB, says, “Nearly 75 per cent of Raven’s die area is covered by standard library blocks and macros. Using these proven IP blocks increased the reliability of Raven while minimising the first-silicon risk.”

Other approaches

Competition is growing and new approaches for remaining relevant are apparent. Silicon alternatives like carbon nanotube transistors are being considered by researchers for further improvement in performance, which may get commercialised in future—once defects and functional problems are dealt with. An example of this is AMD’s EPYC Rome, which uses multi-chip modules along with improved packaging that generates higher yield at a lower cost when compared to Intel CPUs, which use the monolithic die approach.

At the launch of the second-generation AMD EPYC processor, Peter Ungaro, president and chief executive



RV12 RISC-V 32/64-bit CPU core (Credit: <https://roallogic.github.io>)

officer, Cray, said, “As we enter a new era of computing—the exascale era—we are seeing new workloads, new infrastructure, a new way of thinking, and this requires capabilities and performance like we get from our next-generation Shasta supercomputer combined with AMD EPYC processors.”

Design considerations

MCUs are suitable for embedded systems owing to their low cost and small size. These are designed to suit application-specific requirements. For systems like those in the military where real-time performance is critical, MCUs have higher clock speeds, while for simpler applications like microwave ovens, power consumption is lower.

Low power operation with negligible standby current consumption extends product life. Depending on the device, ROM may be permanent or erasable type. MCUs with higher pin count allow additional interfacing. In customisable MCUs, besides microprocessor(s), memory and different peripherals, a digital logic block is also present on the chip.

The 32-bit MCU is gaining popularity due to growing demands for more capabilities. Touch-sensing MCUs, which were not common for simple applications like TVs, are now available at a lower price for the same performance. For achieving high precision, additional features such as

a DC/DC converter and an embedded real-time analysis and diagnostic unit (ERAD) are incorporated at times.

Connectivity standards are an important consideration for extensible and dependable products. Chips designed solely for AI processing need to be connected to host processor(s). In addition to such standards as Wi-Fi, sub-1GHz and Bluetooth Low Energy (BLE), many devices also support others like Zigbee. This gives more flexibility for performing different tasks at the same time.

Security considerations

Any design flaws can be used to steal sensitive information like passwords or private messages. Many MCUs come with security modules. While adding security features, designers need to take care that there is no extra processing burden on the main MCU.

Cryptographic algorithms like advanced encryption standard (AES), secure hash algorithm (SHA-2) and data encryption standard (DES) are generally chosen for secure communication. Once a product is deployed, firmware updates are necessary for security.

Unlike hardware, a patch can be used to deal with software bugs. In software updates, there is no necessity of redesigning hardware. For instance, Intel had to release microcode for its processors such as Intel Xeon and Sandy Bridge, as these were at risk.